MIL-C-45110B (PA)
31 March 1976
SUPERSEDING
MIL-C-45110A (MU)
12 May 1965

MILITARY SPECIFICATION

CARTRIDGE, 90MM, AP-T, M318A1 ASSEMBLING, MARKING AND PACKING

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the assembling, marking and packing for one type of cartridge designated as Cartridge, 90MM, AP-T, M318A1, for use in 90MM Guns, M36, M41 and M54

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550	- Ammunition and Special Weapons: General Specification for
MIL-P-223	- Powder, Black

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions

FSC: 1315

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MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-1168 - Ammunition Lot Numbering
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

7258441 - Gage, Profile and Alignment
7258442 - Gage, Alignment Check
7258443 - Gage, Wear Limit Check
7548467 - Container, Fiber, 90MM, M180A1, Assembly Details and Packing
7548476 - Box, Packing, Ammunition for 90MM Gun Ammunition in Fiber Container, M180A1, Assembly, Details and Packing
9207966 - Cartridge, 90MM, AP-T, M318A1, Assembly

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable specifications and drawings.

3.2 Cartridges.-The cartridges shall comply with all requirements specified on Drawing (dwg.) 9207966, and with all requirements specified in applicable specifications.

3.3 Propellant weight.-The weight of the propellant shall not differ by more than plus or minus 0.25 ounce avoirdupois from that designated by the loading authorization for the particular lot of propellant used.

3.4 Cartridge Case Crimp (Bullet pull).-The cartridge case shall separate from the projectile at a load ranging from 5000 pounds minimum (min.) to 14,000 pounds maximum (max.) when tested as specified in 4.3.1.

3.5 Primers.-The primer shall have the required number of flash holes as specified on the appropriate primer dwg.

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3.6 Moisture content of black powder.-The moisture content of the black powder at the loading station, at the time the igniter assembly is loaded into the cartridge case shall not exceed 0.70 percent when determined as specified in 4.3.3.

3.7 Proving ground.-The cartridge shall comply with the following requirements when tested as specified in 4.3.2.

3.7.1 Muzzle velocity.-The corrected muzzle velocity of the cartridge at ambient temperature shall be not more than 3075 feet per second (f.p.s.) not less than 2925 f.p.s. and the standard deviation of the lot shall be not greater than 12 f.p.s. max.

3.7.2 Tracer functioning.-The tracer shall ignite and there shall be visible evidence of the burning of the tracer composition to the 2000 meter target.

3.7.3 Safety.-There shall be no hangfires or evidence of metal parts separation or breakup in the gunbore or in flight.

3.8 Workmanship

3.8.1 The cartridges shall be free of chips, dirt, grease and other foreign material. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agent.

3.8.2 Care shall be exercised at all times to prevent the rotating bands from being nicked, burred or otherwise damaged.

3.8.3 The painting and marking shall be restored by the loading and assembling contractor on any component furnished to him painted and marked when painting and markings have been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

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4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions

4.2.1 Lot formation.-The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in Standard MIL-STD-105. Changes to the process, specification or drawing, not affecting safety, performance, interchangeability or storage, as determined

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by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded complete round cartridges shall contain:

a. Cartridge cases from one manufacturer in accordance with the same drawing and drawing revision, same specification and specification revision; or, if resized cases are used, from not more than one lot.

b. Projectiles of one lot interfix number from one manufacturer in accordance with the same drawing and drawing revision, same specification and specification revision.

c. Propellant of one lot from one manufacturer.

d. Loaded primers of one lot interfix number from one manufacturer.

e. Loaded tracers of one lot interfix number from one manufacturer.

f. Black powder from one lot.

4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for Critical Defects (and Major Defects, when so specified) shall be 100 percent. Sampling plans for Major and Minor Defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of Major and Minor Defects listed, the Contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each Major Defect and an AQL of 0.65 percent for each Minor Defect. When this method is elected, it shall be applied to all of the Major and Minor Defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

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4.2.2.1 Primer, prior to assembly to cartridge case (see dwg. 8838153 covering a detail of dwg. 9207966)

Categories	Defects	Method of Inspection	Code No.
Critical:			
	1. Any flash hole missing (see 3.5).....	Visual	01001
Major: None defined			
Minor: None defined			

4.2.2.2 Projectile assembly with tracer, prior to assembling cartridge case (see dwg. 9207966)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
	101. Plug and Disc Assembly fails to withstand disassembly torque.....	Torque-Test (CD)	02001
	102. Plug and disc assembly missing damaged or loose.....	Visual-Manual	02002
Minor:	AQL 1.0 percent		
	201. Marking misleading or unidentifiable, including incorrect color code.....	Visual	02003
	202. Evidence of poor workmanship (see 3.8).....	Visual	02004

4.2.2.3 Liner, prior to assembling into cartridge case (see dwg. 8880408 covering a detail of dwg. 9207966)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			

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Minor:	AQL 1.0 percent		
201.	Excessive amount of uncoated surface	Visual	03001
202.	Evidence of poor workmanship (see 3.8)	Visual	03002

4.2.2.4 Cartridge case and liner assembly (see dwg. 8880409 covering a detail of dwg. 9207966).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.5 percent		
201.	Depth of liner from mouth of case	Gage (CD)	04001
202.	Gap between liner and case to extent that propellant can be entrapped	Visual	04002
203.	Evidence of poor workmanship (see 3.8)	Visual	04003

4.2.2.5 Cartridge case prior to loading propellant (see dwg. 9207966)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Primer fails to withstand disassembly torque	Torque-Test (CD)	05001
102.	Exposed liner adhesive tacky or liner not secured to case in area of adhesive	Manual	05002
103.	Primer stake missing	Visual	05003
Minor:	AQL 0.65 percent		
201.	Foreign matter in interior of case (see 3.8)	Visual	05004

4.2.2.6 Igniter assembly (see dwg. 7548695 covering a detail of dwg. 9207966)

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major:	AQL 0.40 percent		
101.	Stitches missing or broken to extent that powder can escape	Visual	06001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	06002

4.2.2.7 Support (see dwg. 8796851 covering a detail of dwg. 9207966)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major: None defined			
Minor:	AQL 1.0 percent		
201.	Material thickness	Gage (CD)	07001
202.	Evidence of poor workmanship (see 3.8)	Visual	07002

4.2.2.8 Cartridge case prior to inserting projectile (see dwg. 9207966)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major;	AQL 0.40 percent		
101.	Igniter assembly, disc or support missing or improperly assembled	Visual	08001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.8)	Visual	08002

4.2.2.9 Cartridge (see dwg. 9207966)

Categories	Defects	Method of Inspection	Code No.
Critical:	100 Percent		
1.	Primer above flush	Gage (CD)	09001

Major: None defined.

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Minor:	AQL 1.5 percent		
201.	Rotating band damaged (see 3.8.2)	Visual	09002
202.	Marking misleading or unidentifiable including incorrect color code	Visual	09003
203.	Exterior coating with an aggregate of bare spots in excess of 1/4 square inch	Visual	09004
204.	Evidence of poor workmanship (see 3.8)	Visual	09005

4.2.2.10 Unsealed fiber container (see dwg. 7548467).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Packing component missing	Visual	10001
Minor:	AQL 1.5 percent		
201.	Glue or asphalt on cartridge	Visual	10002
202.	Cartridge improperly assembled in container	Visual	10003
203.	Cartridge cannot be removed by hand	Manual	10004

4.2.2.11 Sealed fiber container (see dwg. 7548467).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Tape incomplete or badly wrinkled	Visual	11001
102.	Container cut or damaged through all impregnated layers	Visual-Manual	11002
103.	Metal end loose or distorted	Visual-Manual	11003

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Minor:	AQL 2.5 percent		
201.	Tear tab length, min.....	Gage (CD)	11004
202.	Color of tape incorrect.....	Visual	11005
203.	Marking misleading or unidentifiable.....	Visual	11006
204.	Cuts, scuffs or gouges in outer layers.....	Visual	11007
205.	Gap between body and cover of container (exceeding 1/8 inch).....	Visual-Manual	11008
206.	Contents loose in container.....	Manual	11009

4.2.2.12 Sealed wood packing box (dwg. 7548476).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Box damaged exposing contents.....	Visual	12001
102.	DOD symbol missing or incorrent...	Visual	12002
103.	Board loose or broken.....	Visual-Manual	12003
104.	Hardware or strapping missing. broken or loose.....	Visual-Manual	12004
Minor:	AQL 1.5 percent		
201.	Car seal missing, unsealed or improperly attached.....	Visual	12005
202.	Marking misleading or unidentifiable.....	Visual	12006
203.	Handle missing or insecure.....	Visual-Manual	12007

4.2.3 Testing

4.2.3.1 Check test for possible deterioration of primers Major defect, Code No. 01002.-If the total elapsed time between the original acceptance of any primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the primers have been subjected to adverse conditions, however brief, at any time since previous tests, the primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in the applicable specification immediately before the primer lot is assembled into the cartridges. (see 6.4).

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4.2.3.2 Moisture content of black powder (see 3.6) Major Defect - Code No. 06003.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. The contractor shall test black powder from at least one sample igniter assembly from each eight hours of complete round production. The black powder shall be tested as specified in 4.3.3 and any failure to comply with the applicable requirements shall be cause for rejection of the eight hours complete round production represented by the sample.

4.2.3.3 Propellant weight (see 3.3) Major defect - Code No. 09006.-The propellant shall be weighed, identified and check-weighed 100 percent. The check-weighing shall be accomplished independently of the original weighing and, if performed manually, shall be performed by another operator. Any propelling charge which fails to comply with the applicable requirement shall be classed defective and shall be returned for correction and resubmission. Equipment shall be in accordance with 4.2.4.

4.2.3.4 Profile and alignment (see dwg. 9207966) Major defect - Code No. 09007.-The cartridges shall be gaged 100 percent for profile and alignment. Any cartridge which fails to meet the applicable requirement shall be classed defective and be removed from the lot. Equipment shall be in accordance with drawing 7258441.

4.2.3.5 Cartridge Case Crimp (Bullet pull) (see 3.4 and dwg. 9207966) Major defect - Code No. 09008.-The sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-414, Section B, Part II, Table B-3, Code letter F, and an AQL of 1.5 percent. The test shall be performed as specified in 4.3.1 with contractor designed equipment.

4.2.3.6 Proving Ground.-The following tests shall be performed as specified in 4.3.2. These tests shall be conducted at a Government proving ground in accordance with the Acceptance Test Procedure CR-90G-1J. The sample cartridges shall be fired alternately with an equal number of calibrated reference rounds in a gun for which the cartridge is standard and observation made for conformance to all applicable requirements.

4.2.3.6.1 Muzzle velocity (see 3.7.1) Major defect - Code No. 09009.

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4.2.3.6.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-414, Section B, Part II, Table B-3, Code letter I and AQL 1.0 percent. In addition, if the standard deviation of the sample cartridges multiplied by the factor 0.75 exceeds the lot requirement, the lot shall be rejected.

4.2.3.6.1.2 After three consecutive lots have met the criteria of 4.2.3.6.1.1, the sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-414, Section B, Part II, Table B-3, Code letter F, and AQL 1.0 percent. In addition, if the standard deviation of the sample multiplied by the factor 0.65 exceeds the applicable lot requirement, the lot shall be rejected.

4.2.3.6.2 Tracer functioning (see 3.7.2) Major defect, Code No. 09010.

4.2.3.6.2.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sample cartridges fired in the test for muzzle velocity (see 4.2.3.6.1.1) shall be observed for tracer functioning. If three or more sample cartridges fail to comply with the applicable requirements, the lot shall be rejected.

4.2.3.6.2.2 After three consecutive lots have met the criteria of 4.2.3.6.2.1, the sample cartridges fired in the test for muzzle velocity (see 4.2.3.6.1.2) shall be observed for tracer functioning. If two or more sample cartridges fail to comply with the applicable requirements, the lot shall be rejected.

4.2.3.6.3 Safety (see 3.7.3).-The cartridges fired in the test for muzzle velocity shall be observed for conformance with the applicable requirements. Failure of any cartridge to meet the requirement shall be cause for rejection of the lot. The final disposition of any lot, having cartridges with special defects, shall be made in accordance with paragraph 6.6. Safety defects shall be classified as follows:

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4.2.3.6.3.1 Special defect, Code No. 09011 (see 6.6).- Separation or breakup of windshield from projectile.

4.2.3.6.3.2 Critical defect - Code No. 09012.-Breakup of any components (excluding windshield) in the gunbore or in flight, or hangfires.

4.2.3.6.4 Extreme Temperature Firing (For Information Only) - Code No. 09013.-Twenty (20) cartridges shall be selected from each of the first three production lots and fired for muzzle velocity, target impact dispersion and maximum chamber pressure information at -35 degrees Fahrenheit (°F) (10 rounds) and 125°F (10 rounds). Tests shall be conducted as specified in 4.3.2.3.

4.2.3.6.5 Target-Impact-Dispersion (For Information Only).- The same cartridges used in the test for muzzle velocity (See 4.2.3.6.1.1 and 4.2.3.6.1.2) shall be observed as specified in 4.3.2.1 for target-impact-dispersion information.

4.2.4 Inspection Equipment.-The inspection equipment required to perform the examination and test prescribed in this section is identified under the "Method of Inspection" headings in 4.2.2 and in 4.2.3 (See 6.3).

4.3 Test Methods and Procedures

4.3.1 Cartridge Case Crimp (Bullet Pull).-The cartridge shall be tested in an appropriate tensile testing machine to determine the security of the crimp of the cartridge case to the projectile. The load shall be applied in an axial direction with the rate of travel of the testing machine head from 0.100 to 0.125 inch per minute.

4.3.2 Proving Ground

4.3.2.1 Velocity and Target Impact Dispersion.-The ammunition fired in this test shall be conditioned at a temperature of 70°F $\pm$ 2°F in a test chamber until all propellant is stabilized at a uniform temperature prior to firing. This normally requires a period of 24 hours. The cartridge shall be fired at a cloth target placed at a distance of 2000 meters from the muzzle of the gun. Vertical and horizontal hit pattern results shall be expressed as

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probable errors in mils and shall be recorded on the firing record for informational purposes only (See 6.5). Muzzle velocity shall be determined and recorded for each round.

4.3.2.2 Tracer Functioning.-Cartridges fired during the velocity test shall be observed for tracer functioning to the 2000 meter target. The distance over which the tracer burns shall be recorded to the nearest estimated 500 meters (i.e., failure, 500 meters, 1000 meters).

4.3.2.3 Extreme Temperature Firing (Information Only).-For this test, ten (10) rounds shall be temperature-conditioned at -35°F ± 2°F and ten (10) rounds shall be temperature-conditioned at +125°F ± 2°F until all propellant is stabilized at a uniform temperature prior to firing. This normally requires a period of 24 hours. The cartridge shall be fired at a cloth target placed at a distance of 2000 meters from the muzzle of the gun. Muzzle velocity, maximum chamber pressure and target impact dispersion data shall be recorded.

4.3.3 Moisture content of black powder.-The moisture content of the black powder shall be determined in accordance with procedure specified in Specification MIL-P-223.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level A.-The containers shall be packed in accordance with dwg. 7548467.

5.2 Packing

5.2.1 Level A.-The containers shall be packed in accordance with dwg. 7548476.

5.3 Marking.-The packing container shall be marked in accordance with dwgs. 7548467 and 8796521. Marking of the packing box shall be in accordance with dwgs. 7548476 and 8796522.

6. NOTES

6.1 Ordering Data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data Cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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6.2 Inspection Code Numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3, should he elect to do so.

6.3.1 SIE Designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing, as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government, and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with

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SIE designs if not otherwise covered thereby (See 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Cost of Check Test.-The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the assemblies or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage, etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.5 Submission of Test Data and Ammunition Data Cards.-In addition to the normal distribution of records, when the cartridge is procured by the Army Armament Command, one copy of all ballistic test data and ammunition data cards shall be forwarded to each of the following:

Commander, Picatinny Arsenal
ATTN: SARPA-QA-A-A and
SARPA-AD-E-A
Dover, New Jersey 07801

6.6 Special defects

6.6.1 Definition.-A special defect is a defect that judgement and experience indicate could, depending on the degree of variance from the design requirement, result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product.

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6.6.2 Disposition.-Upon investigation and analysis of lots exhibiting special defects by the responsible technical activity a recommendation of acceptance on waiver or rejection shall be made to Director of Quality Assurance, U.S. Army Armament Command. Final acceptance of such lots on waiver shall be made only by the Director of Quality Assurance, U. S. Army Armament Command.

6.7 Proving Ground Test

6.7.1 Test Validity.-If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test shall be performed with additional samples.

6.7.2 Test Summary: See next page.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1315-A-096

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PROVING GROUND TEST SUMMARY

TEST	SAMPLE SIZE INITIAL (NOTE 1)	SAMPLE SIZE SUBSEQUENT		REQUIREMENTS	SAMPLING PARAGRAPH	TEST PROCEDURES PARAGRAPH
Muzzle Velocity	25	10	a.	Muzzle Velocity (3.7.1)	4.2.3.6.1	4.3.2.1
			b.	Tracer Functioning (3.7.2)	4.2.3.6.2	4.3.2.2
			c.	Target Impact Dispersion (Info)	4.2.3.6.5	4.3.2.1
			d.	Safety (3.7.3)	4.2.3.6.3	4.3.2.1
Extreme Temperature Firing	20	0	a.	Pressure (Info)	4.2.3.6.4	4.3.2.3
			b.	Muzzle Velocity (Info)	4.2.3.6.4	4.3.2.3
			c.	Target Impact Dispersion (Info)	4.2.3.6.4	4.3.2.3
			d.	Safety (3.7.3)	4.2.3.6.4	4.3.2.3

NOTES: 1. Until three (3) consecutive lots are accepted.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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